

The Fifth Indian Voyage to Antarctica

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Expedition Leader

Abstract

The Fifth Indian Scientific Expedition to Antarctica left India on 30th Nov., 1985 and reached the icy continent on 24th Dec., 1985. It started its return journey on 3rd March, 1986 after completing all its objectives. This paper gives a brief account of how the expedition was conducted and the tasks that were achieved, both in the scientific and logistic categories. The expedition left behind a team of fourteen members of the Third Wintering at Dakshin Gangotri and brought back the thirteen member Second Wintering Team.

Introduction

After four successful Indian expeditions to Antarctica, the Fifth Expedition sailed on 30.11.85. The objectives of the expedition were the continuation of India's scientific programme in Antarctica, with a major thrust on taking the activities further into the interior of Antarctica than hitherto attempted. In addition to this scientific programme, the task assigned to the expedition was the repair and maintenance of the existing permanent structures of the Station at Dakshin Gangotri and Maitree camps.

The Team

The 88 member team was drawn from several different scientific institutions of the country, and from the three wings of the Defence Services, namely Army, Navy and Air Force for the logistic support. The organisations which participated in the expedition are as follows:—

- Geological Survey of India (GSI)
- National Geophysical Research Institute (NGRI)
- India Meteorological Department (IMD)
- National Institute of Oceanography (NIO)
- Physical Research Laboratory (PRL)

National Physical Laboratory (NPL)
 Indian Institute of Geomagnetism (IIGM)
 Bharat Heavy Electricals Ltd. (BHEL)
 National Remote Sensing Agency (NRSA)
 Defence Research and Development Organisation (DRDO)
 Oil and Natural Gas Commission (ONGC)
 Indian Institute of Technology, Bombay
 Indian Institute of Technology, Madras
 Indian Army
 Indian Navy
 Indian Air Force
 Films Division
 Department of Ocean Development (DOD)

The scientific team had twenty-three members representing thirteen different scientific organisations. The Army had a twenty-six member team and the Navy and Air Force were represented by eighteen members each. There was a representative each from the Department of Ocean Development (DOD) and the Films Division of Government of India. One workman from DRDO was also included in the team. The Armed Forces Medical Corp included a surgeon, a physician and an anaesthetist.

Loading Operations

The ship chartered for the Expedition was M.V. THULELAND, a Swedish iceclass IA, Super hull. The ship has six holds and two single and one twin crane for handling cargo, capable of lifting 25 tonnes upto a height of 26 metres. During the expedition the vessel accommodated 88 charter's personnel in crew cabins, on the poopdeck, maindeck and container accommodations. The uniform time charter executed between the owners and the charterer was strictly followed both by the crew and expedition members. The team, after reporting itself at Marmagao port, Goa, started the loading operations almost round the clock and 650 tonnes of cargo were distributed as under:-

	Cargo	Tonnage (MT)
Holds	Fuel	
	Other cargo	575
	Aircrafts.	
Deck	Deck cargo	75

Fuel was stored in the ship's tanks with a capacity of 14000 litres each and thus, seven tanks were filled. The rest of the fuel was kept in the tanks supplied by DRDO and in sealed barrels of the Indian Oil Corporation. The ship's tanks and DRDO containers were thoroughly cleaned and flushed before, filling. The other cargo consisted of the construction material, spares of vehicles, clothing, camping equipment, food (from FTRI, DFRL, Super Bazar, MAFCO and Modern Food Industries), hydrogen gas cylinders, medical supplies, L.P.G. cylinders, scientific equipment, communication equipment etc.

The Cruise — Destination Antarctica

The ship left Marmagao port on 30.11.85 at 1740 hrs. IST for Antarctica. Some deviation in the route had to be made only at Mauritius for 5^{1/2} hours on 8.12.85 for changing a few crew members. The ship did not berth alongside and the crew was changed by a tugboat. The journey from Mauritius to Antarctica faced a phase of difficult weather conditions, apart from the usual turbulent sea in the roaring forties. On 15.12.85 a cyclonic storm with a central pressure of 968 mb was observed at location Lat. 60°S — Long. 15°E. The ship experienced very rough weather with wind speeds of 55 knots. On 16.12.85 the storm was faced on the way to position Lat. 62°S — Long. 35°E. It was further discovered by the expedition leader and the captain of the ship that if the ship was allowed to proceed along its pre-determined course, a storm with a speed of 80 to 100 knots would be encountered, as per the facsimile charts. This could be dangerous for the expedition and detrimental to the cargo, helos etc. Hence a westerly course was maintained upto 19.12.85, till the storm passed away and the ship's course was accordingly again altered towards its destination. Weather charts were regularly received during the cruise.

During the cruise, regular meetings were held by the Leader with the scientific and logistic team members, individually and collectively, to formulate plans and programmes for unloading of cargo, shifting of materials, scientific work and execution of other jobs during the journey and after reaching the continent. Regular recreational facilities such as films, books, cards, table tennis, chess, carrom, gymnasium etc. were kept at the disposal of the members. Several committees were formed by the Leader to look after the mess, recreation, watch and ward, and stores. The warm clothing was issued to each member before entering the cold zone.

Arrival

The ship entered the pack ice on 17.12.85 and the latest facsimile chart was specially requested from Molodezhnaya which was obtained on 19.12.85, with two recommended routes plotted on it, based upon the concentration of pack ice. The ship altered its course to converge on the nearest point of the recommended routes. A reconnaissance flight by the helicopters helped in finding favourable leads which were followed and the ship reached polynya without any further

difficulty. It arrived close to the fast ice on 24.12.85, which was the Christmas eve. Immediately on arrival at fast ice, the first sortie was organised for landing at Dakshin Gangotri Station and the landing was made at 1740 hours local time, to the jubilation of the members of Second Wintering Team anxiously awaiting our arrival.

Shifting the Loads at Antarctica

As per the action plan already chalked out, unloading operations started immediately on the morning of 25.12.85. Initially the cargo lying on the deck was unloaded, out of which priority was given to the scientific cargo. The very first sortie was taken for dropping scientists and their instruments at Dakshin Gangotri to enable them to start scientific work without losing any time. Having completed the transfer of scientists, essential scientific equipment and other related cargo on 25.12.85, another scientific team was shifted to Maitree camp along with instruments and infrastructural facilities on 26.12.85 after activating the camp one day earlier i.e. on 25.12.85. On 27.12.85 a fully operational communication system was set up at Maitree.

Due to intermittent bad weather the ship could not come alongside till 12.1.86. But due to recurrence of bad weather, again it had to move away from the shelf. It was only from 16.1.86 that the ship could stay alongside for a few days, with breaks, till 20.2.86. The dates on which the ship was alongside are as follows:-

From (Local time)	To (Local time)
12.1.86 1115 hrs.	12.1.86 1940 hrs.
16.1.86 1030 hrs.	23.1.86 2325 hrs.
24.1.86 1530 hrs.	24.1.86 1030hrs.
04.2.86 1020 hrs.	04.2.86 1720 hrs.
05.2.86 1050 hrs.	11.2.86 2045 hrs.
13.2.86 1530 hrs.	14.2.86 1310 hrs.
17.2.86 1530 hrs.	20.2.86 1000 hrs.

These periods were utilised to the maximum extent possible for transferring the loads from the ship to the station on piston bullys and for timely transportation of the cargo. Simultaneously, cargo was shifted in the helicopters also, avoiding any flight on the days of bad weather. The break up of the cargo shifted from the ship using air and ground transport during the above period is as follows:-

Cargo	By helicopters (MT)	By sledges (MT)
1. Fuel	72	108

2. Construction and allied stores	80	45
3. Spares and allied stores	15	15
4. Food stuff	10	50
5. Hydrogen/L.P.G cylinders	2	13
6. Scientific and communication stores	30	—
7. Containers	—	7
Total	209	238

The heavier cargo such as the fuel containers with a capacity of approximately 4500 litres each, food containers and articulating sledges were shifted much in advance to ensure timely and proper positioning of the fuel and stores for the Third Wintering Team.

Helo Operations

The helicopter flying operations commenced on the day the ship reached Antarctica i.e. on 24.12.85 and terminated on 03.3.86, when the ship departed from Antarctica. The helicopter operations were carried out for logistic support to scientific teams in various camps, personnel and cargo shifting, photo reconnaissance and aerial reconnaissance, ice-berg landing, casualty evacuation and medical support and other operations in hilly, rough and icy terrains, including that for the identification of camp site in the interior area of Petermann Range for future work. These were also utilised for route reconnaissance to aid in ship's navigation in fast ice and pack ice during both downward and upward voyages. A summarised account of the flying operations by both Navy and Air Force is given in the following table.

	Navy (Two Chetaks)	Air Force (Two MI8)
Passenger ferrying (Nos.)	608	1369
Load (tonnes)	14	248
Flying (hours)	274	117
Sorties (nos.)	258	321

Scientific Programme

The main thrust of the Expedition was on the extension of multidisciplinary scientific activities into the more interior areas of Antarctica besides expansion of the activities already in progress. Accordingly, the scientific team was divided

into four major groups. It was ensured that each group was multidisciplinary in nature and logistically self contained. The following scientific disciplines were tackled from the respective camps:

Gruber camp (Wohlthat mountains)	: Geology, glaciology, remote sensing, biology, performance appraisal of clothing.
Maitree camp (Schirmacher hills)	: Geophysics, geology, glaciology, biology, remote sensing, non-conventional energy, meteorology, materials research, experiments on indoor horticulture.
Dakshin Gangotri Camp	: Meteorology, geomagnetism, radio-physics, glaciology, non-conventional energy, remote sensing on shelf ice, materials research, experiments on mineral tablets and indoor horticulture.
Polynya (Ship)	: Biology and oceanography, meteorology.

Besides the above scientific disciplines, during the Expedition, two studies were initiated for the first time in the Indian programme, viz. search for meteorites and reconnaissance aerial photography of the mountain areas. Extensive search for meteorites was made both on ground and by low flying from helicopters, but no meteorite could be located. A short note on the search during this expedition is given as Annexure 'A'. Also, aerial photography was carried out over Gruber massif and Petermann area by devising a platform in a Chetak helicopter and fitting an Agiflite camera onto it. Though the results are encouraging, all of them are not found to be completely fit for stereoscopic interpretations. A brief note on this reconnaissance is given in Annexure 'B'.

It was for the first time that a camp was established at Gruber massif in the eastern flank of the Wohlthat mountain chain. The camp was provided with the services of a doctor and a wireless communication set. The wireless communication could link this camp with the Maitree camp and the main station. This camp site lying about 180 kms inside from the shelf is one of the windiest places in Antarctica, often resulting in the tents blowing off due to high winds. No container accommodation was till then available for such camps.

At Maitree, the camp was activated on the next day of arrival at Antarctica i.e. on 25.12.85 and the scientific component was shifted on 26.12.85. A doctor and a communication team was included in the group. The camp was closed on 19.2.86, after fifty-six days of stay by the scientists and the logistic teams.

At Dakshin Gangotri station the scientific team was landed on the very next day of arrival. Initially the scientists were accommodated in tents which got heavily damaged and buried in snow only after two days. The scientists were then accommodated in the DRDO complex accommodation and hangar

garage where infra-structural facilities like provision of water, electricity and laboratory facilities were arranged and improvised. The camp was closed on 28.2.86 after a stay of sixty-six days, thus making maximum time available for recording observations.

A modular laboratory was installed on the ship before departure from Goa for carrying out experiments on board the ship during the cruise and while at Antarctica. A hydrographic winch installed on the ship catered to the collection of marine samples on board the ship. The oceanographic group used the modular laboratory and the winch.

Constructions and Maintenance Work at the Station

After the construction loads were shifted from the ship to Dakshin Gangotri Station, the construction work, which included the strengthening of the station building and routine maintenance, was immediately started. The following tasks were accomplished:

1. Strengthening of link block and fuel tank room from inside.
2. Construction of staircases in blocks A & B.
3. Extension of all generator exhausts, general purpose exhaust and fresh air intakes.
4. Extension of snow melt tank.
5. Clearance of snow from blocks A & B and between them.
6. Construction of gable roof over toilets and Sauna.
7. Construction of cabin in Hangar for alternate communication room.
8. Construction of base for SATCOM in block B.
9. Strengthening of gables in blocks A & B.
10. Construction of new floor, strengthening of beams and establishment of two generators in the station garage.
11. Strengthening of roof in block A.
12. Shifting of SATCOM system from DRDO complex to alternate communication room in the Hangar garage.
13. Shifting of SATCOM dome from link block to new base in Block B.
14. Construction of generator cabin in vehicle Hangar and establishment of one generator there.
15. Digging of two Portakabins from DRDO complex and their installation in Hangar as alternate accommodation.

16. Repair, maintenance and activation of the Maitree huts and their essential services.
17. Modification of sewage system of toilets at Maitree camp and construction of a generator cabin.
18. Construction of summer toilets at Dakshin Gangotri immediately on arrival of the summer team.
19. Repair, maintenance and servicing of vehicles, snow cutters, cranes, generators, boilers, snow melt plant at Dakshin Gangotri.
20. Servicing of refrigeration plant and other electrical systems at the station.
21. Strengthening of roof and floor of DRDO Complex, provision of a new escape shaft and maintenance of living and kitchen modules of the complex.

Communication Systems

At Dakshin Gangotri Station two SATCOM terminals are available. One is in the main station building with telephone, telex, fax and SSTV facilities and the other is in the main garage with telephone and telex services. The spare SATCOM was non-functional because of a defective power supply unit. This power supply unit was replaced during the Expedition and the system was made fully operational and serviceable. The performance of the spare SATCOM was confirmed by repeated tests and it is now fully operational.

Arrival India

After having successfully completed all the allotted tasks at Antarctica, the Expedition sailed back for India in the early hours of 03.3.1986, bringing along with the members of the Second Wintering Team and leaving behind a fourteen member Third Wintering Team. The expedition landed safely back in India at 09.00 hrs on 24.3.1986.

Some Highlights of the Expedition

In its quest to achieve the identified objectives the Fifth Indian Scientific Expedition to Antarctica had many accomplishments some of which are as follows:

1. A record stay of 69 days on the icy continent was made, making the fullest use of the available summer time. Members stayed for 56 days at MAITREE Camp and for 66 days at DAKSHIN GANGOTRI for completing the assigned tasks.
2. Positioning of the scientific and logistic teams was done at respective camps within the shortest possible time after the arrival of the ship in Antarctica. Within two days of the arrival, the scientific work was in full swing.

3. Extension of multidisciplinary scientific studies into the interior of the continent, by establishing additional summer camp in Wohlthat mountains was achieved. Wohlthat mountain chain is one of the highest and difficult mountain chains girdling Antarctica.
4. Geological mapping and regional mineral assessment of the entire Gruber massif of the Wohlthat range and the northern part of the Petermann range was completed. Examination of the geology of nine NUNATAKS, located farther away from the Schirmacher hills was also carried out.
5. Geophysical survey of the entire Schirmacher hills for identifying the probable loci of mineralisation was accomplished.
6. Reconnaissance aerial photography of Gruber massif and Petermann range of Wohlthat mountains, supported by ground truths, to help develop technique for the interpretation of aerial photos of the area was achieved.
7. Aerial profiling for remote sensing over Gruber massif and Petermann range of Wohlthat mountain chain was carried out. The four band spectral signatures and temperatures have been recorded, which will help in the interpretation of satellite imagery of these areas.
8. Establishment of fully operative communication systems at Wohlthat and Maitree camps, during the period of stay was achieved.
9. Studies on the environmental preservation and strict observance of rules for maintaining the Antarctic ecosystem were fully enforced.
10. Launching of a record number of sorties for the support to scientific teams and shifting of loads by the Navy and Air Force, free from any accident was achieved. The Navy flew sorties for an aggregate of 274 hours and Air Force for 117 hours.
11. A small helipad for landing MI-8 helicopters was established in the interior of Petermann range, thus opening avenue for temporary camp site for future summer expedition parties.
12. Experiments on growing of vegetables at Dakshin Gangotri and Maitree were conducted.
13. Repairs, maintenance, servicing and strengthening of the station much before the targeted date were completed.
14. Established a Data Collection Platform at Dakshin Gangotri which transmits directly the weather data to India through the satellite.
15. Positioning of the Third Wintering Team was completed with all infrastructural facilities at the station before the scheduled date, thus enabling them to initiate scientific work well in advance.

Some of the interesting aspects and activities of the expedition are depicted in Fig 1 to 20.

Reference

- KAUL, M.K (1986) :Preliminary report — Fifth Indian Scientific Expedition to Antarctica (1985-86)
Department of Ocean Development, Govt, of India.

The Fifth Indian Voyage to Antarctica



Fig. 1. The Expedition ship M.V. Thuleland alongside the fast ice.



Fig. 2. Establishment of summer camp in the vicinity of Dakshin Gangotri station.



Fig. 3. The summer camp buried under snow after a severe blizzard within two days of establishing it.



Fig. 4. The 'beard' after the 'blizzard'.



Fig. 5. snow accumulation around the station after the blizzard.



Fig. 6. The summer camp with communication facility at Gruber massif.



Fig. 7. Extension of shafts and exhausts in the main building at Dakshin Gangotri.



Fig. 8. Part of the exposed section of the station after snow clearance.



Fig 9. Strengthening of roof of the DRDO complex at Dakshin Gangotri and construction of a new escape shaft.

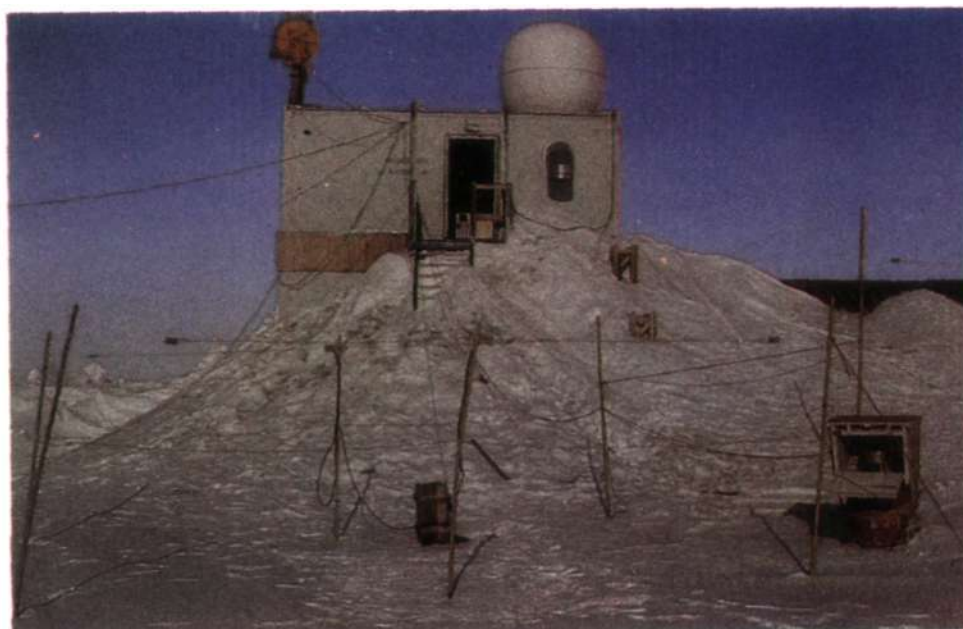


Fig. 10. Establishment of alternate SATCOM communication system at Hangar garage site.



Fig. 11. The new position of the SATCOM dome in block B



Fig. 12. Reconnaissance for locating a suitable camp site in the Woblbat ranges, west of Gruber massif.

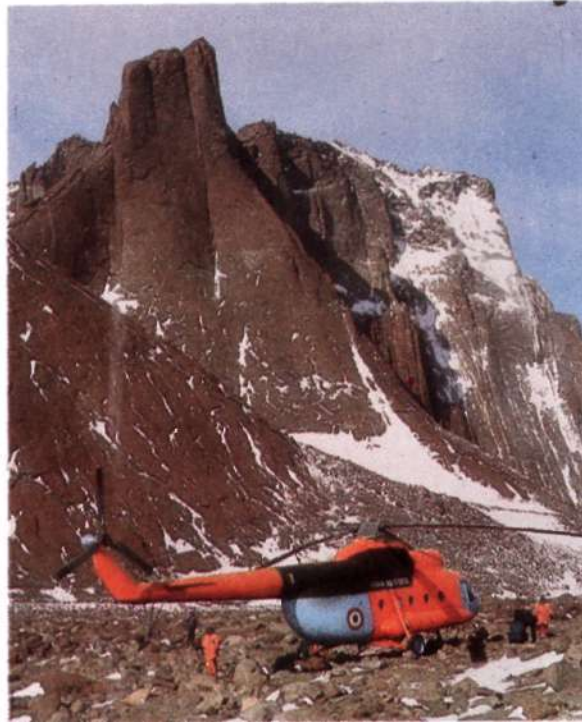


Fig. 13. Clearing the ground for the camp site in Petermann ranges.



Fig. 14. Geological mapping around Nunataks.

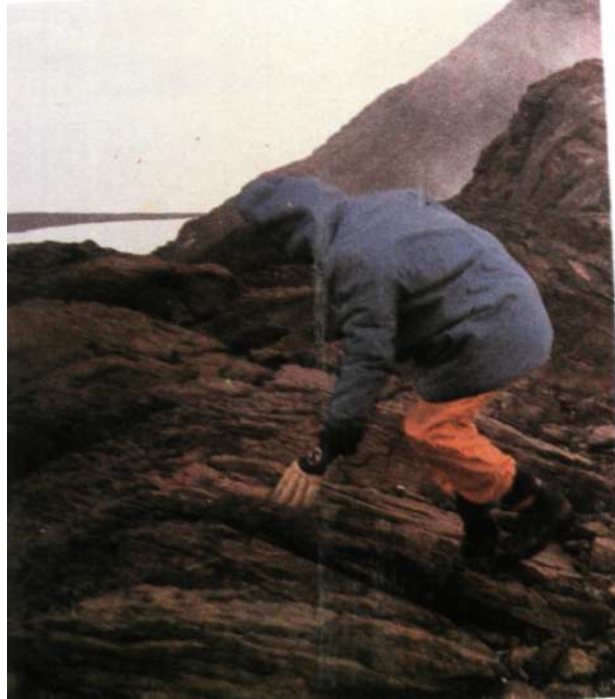


Fig. 15. Negotiating a cliff in course of geological mapping.



Fig. 16. Fixing the position during geological traverses in the Wohlthat mountains



Fig. 17. Geological mapping in the higher reaches of eastern Wohlthat range.



Fig. 18. A stranded iceberg in fast ice off Daksbin Gangotri shelf.



Fig. 19- The friendly 'natives'—Emperor penguins watching from the fast ice.



Fig. 20. The fourteen member third wintering team, which stayed back, seen with the expedition leader on the eve of departure of the Fifth Indian expedition back to India

Search for Meteorites During the Fifth Antarctic Expedition

- Background:* Antarctica has proved to be the most attractive field for meteorite search. Since the discovery of the first meteorite in December 1912 by the Australian expedition party at Adelie Land, there has been a continuous search for meteorites in other areas of the continent. This effort has led to the large collection by Japanese and American expeditions near Yamato mountains and Allen hills, respectively. India's interest in search for meteorites in the area of Queen Maud Land, where the Indian scientific activity is going on at present, was all the more stimulated by the past collection of a meteorite in two fragments from this very area by a U.S.S.R team in 1961.
- Common types:* The three common types of meteorites which fall on earth as extra-terrestrial bodies are the stony meteorites (representing 93% of meteorite fall), iron meteorites (representing about 6%) and stony-irons (representing about 1%), which are the tools for understanding the origin and evolution of the solar system. Few meteorites have also been traced to Moon and Mars as their parent and many still unknown facts may be buried under the ice of Antarctic continent.
- The search programme:* It was in the light of the above that the decision was taken during the DOD - ADCOS meeting on April 17, 1985 that a meteorite search programme should be launched by India also and should be initiated during the Fifth Indian Expedition to Antarctica. The meteorite search programme being a part of the total geological and glaciological programme of Geological Survey of India (GSI) and Physical Research Laboratory (PRL), it was decided that due to logistic constraints regarding helicopter sorties etc., this programme will be clubbed with geological and glaciological programmes, alongwith supporting programmes on meteorology and radiosience. The meteorite search programme was, therefore, initiated at Antarctica during the Fifth Expedition as a GSI - PRL collaborative programme. The participants in this programme were M.K. Kaul, V.N. Nijampurkar, R.K. Singh, D. Srivastava, S. Jayaram and S. Mukerji.
- Probable loci:* Several distribution and accumulation models for meteorites have been worked out by virtue of actual field experience of different field workers. The most common and simple model is that when a meteorite falls on bare ice it sinks into the

upper softer layers and gets trapped in the solid ice. The latter, being constantly in a state of flow, carries the meteorite along its flow trajectory till an obstruction in the form of a mountain barrier or a nunatak folds the lower ice layers in an upward direction. Due to ablation by wind or melting, the ice layers thus folded get exposed and along with them the meteorites concentrate. Any flow trajectory that does not meet an obstruction may ultimately reveal itself in the outermost margins of the continent or on shelf ice-polar ice transition zone. Hence in addition to these probable loci of meteorites, the blue ice zones, terminal moraines near the obstructions and isolated stone like bodies standing out conspicuously in a white expanse of ice were also earmarked for examination. Such an exercise of delineating the probable loci was done on satellite imagery and available maps before the departure of the expedition.

*Field search
and collection:*

Two types of search were conducted for locating any meteorite during the Expedition:-

1. During routine geological and glaciological traverses on the ground for geological and glacio-geomorphological mapping, attention was paid to any unusual looking rock or stone of different shapes. Attention was paid especially to any rock that would show depressions called 'thumb prints', fusion crusts or chondrules. Any 'suspects' were thoroughly examined under a hand lens. These were examined for specific gravity with reference to the surrounding debris. In spite of negative tests some possible 'suspects' were still collected and brought back to India. They were handled with utmost care being photographed insitu and packed in plastic bags avoiding contamination. No 'suspect' was broken, hammered or subjected to any impact. Before packing, these were properly numbered (Fig. 21). The 'suspects' thus collected were located in the Gruber and Schirmacher moraines as black heavy chunks.

2. In addition to the above ground search, an aerial search was carried out in the area from south of Gruber massif (midway to Payer mountains) to Schirmacher hills in the regions of suspected meteorite occurrence. Also the shelf area was surveyed aerially to locate any conspicuous object within the ice. Flying was carried out by Chetak helicopters at low altitudes varying from 100 to 500 feet and covering an area of about 5000 sq kms. All the members were aided by binoculars during aerial search. All the helicopter pilots and some other members were briefed about the background

information on meteorites. A few guidelines and characteristics of the meteorites were explained and distributed to them in the form of a note by the Leader. The fliers were very conscious of the meteorite search programme whenever they used to fly and were very watchful.



Fig. 21. The meteorite suspects' as collected in the field.

Results:

The 'suspects' collected during the above search were subjected to chemical analyses and proved to be ilmenite chunks and not any meteorite. The ore microscopic and other petrological studies revealed that the predominant ore mineral is ilmenite varying from 90% (visual estimate) to less than 50% in different samples. Higher percentages of titanium are noted in samples showing massive ilmenite while lower percentages are noted in samples that show admixing with thin veins or isolated patches. A typical photomicrograph of the polished section of one of the samples is given in Fig. 22.

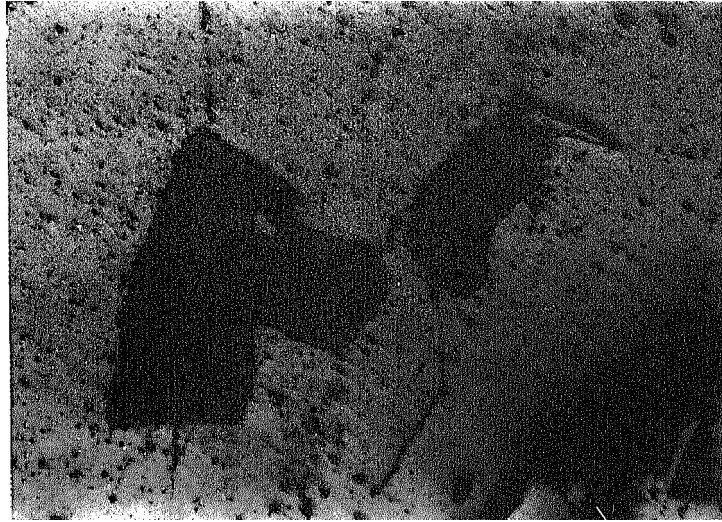


Fig. 22 Photomicrograph of the ilmenite sample showing magnetite crystals (negative scale is 6.3×10).

Suggestions:

It is suggested that the search for meteorites should be continued in the ensuing expeditions and the scientists not familiar with the meteorites and their characteristics should be exposed to the gallery of meteorites collections in the G.S.I museum, Calcutta to enable them to easily identify a 'suspect' in the field. They should also be apprised with the field identification techniques for meteorites at GSI and PRL laboratories.

Reconnaissance Aerial Photography

1. In the absence of any vertical air photographs, pertaining to the Wohlthat mountains with the Indian expedition, it was decided during the Fifth Antarctic Expedition to attempt, on an experimental basis, the vertical air photography of a part of this mountain chain. The platform used was a Chetak helicopter of the Indian Navy in which an Agiflite camera was mounted vertically. The contraption was devised in such a manner so as to save the camera lens from fogging due to exposure while flying at high altitudes.
2. The photography was carried across the trend of the main mountain range along a pre-determined path. The flight lines are shown in Fig. 23. Each

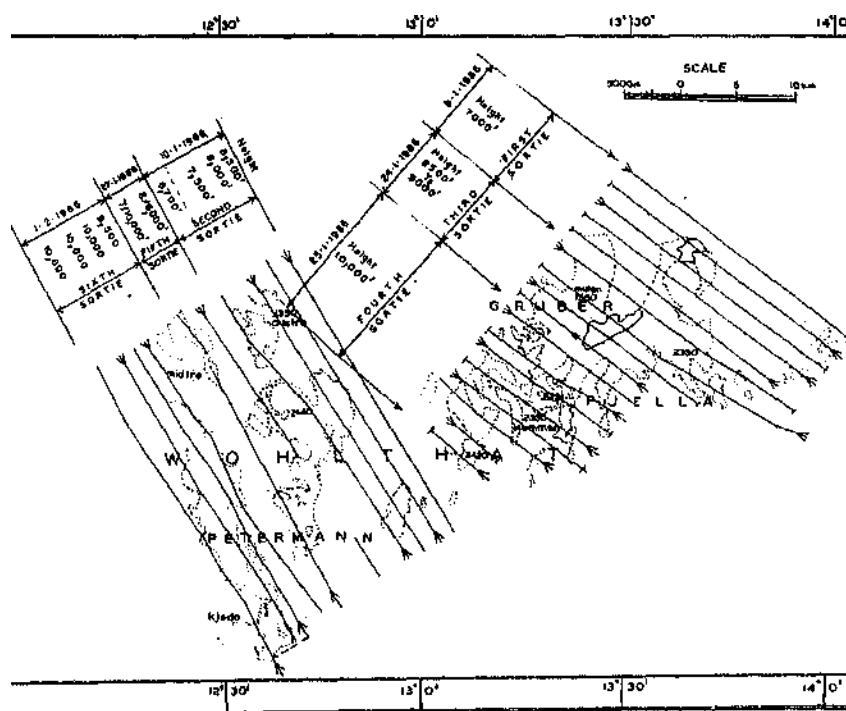


Fig. 23. Map showing flight lines and height of flying for reconnaissance aerial photography in the eastern Wohlthat mountains during the Fifth Indian Antarctic Expedition.

frame was exposed in such a manner so as to affect an overlap on the next frame along the flight line. After completion of one flight line, the return flight was taken on a parallel line, but offset laterally, to an extent permitted by flying constraints. As such the lateral overlap was not possible though the linear overlap could be controlled along the line of flight.

3. Though the photography was done on an experimental basis, fully realising the constraints of the platform used, it was tried, as far as possible, to maintain a uniform height of flying along a particular traverse line to later work out the scale of photograph reasonably accurately. Due to eddy effect of the wind and other flying constraints, the uniformity, however, could not be maintained.
4. After return of the Expedition, the films were developed and photographs enlarged. The clarity of the pictures was found to be remarkably satisfactory (Fig. 24). An example of a stereo-pair is reproduced in Fig. 25. The



Fig. 24. Aerial photograph showing distinct boundaries between moraines (a), wind blown snow (b), surface ice melt (c) and polar ice (d).

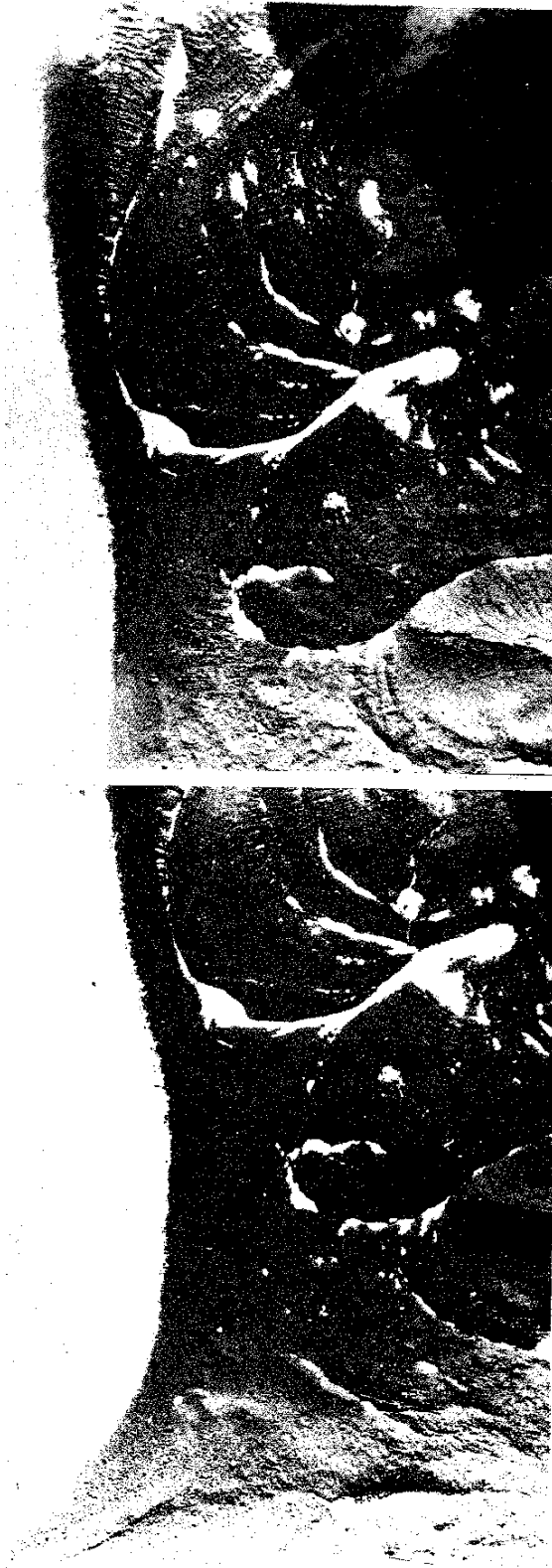


fig-25. A stereo pair of lobate (ice-cored) moraines forming interesting geomorphological feature in the Petermann ranges. Photointerpretation reveals elevated terminal cliffs.

photographs can be used by the subsequent expeditions for limited ground checks and delineation of detailed features of geological and geomorphic interest, including fixation of the scale of the photographs, in those stretches which have been conveniently depicted on the photographs. The overlap, however, is linear only in few cases and the scale would differ from flight to flight due to changes in flying altitudes.

5. .It is suggested that in view of the clarity of results and convenient linear overlap, wherever obtained, during the above exercise, a detailed plan for aerial photography could be worked out for some stretches of Wohlthat mountains where technical flying difficulties of helicopters is minimum. The flight lines can be oriented and extended in such a way as to allow lateral overlap also which would enhance the utility of photographs. Attempts at maintaining uniform heights of flying on the flight lines can be made. One more exercise of flying with this objective is worth making to obtain quick look prints, within the limitations, as an aid to surveying on the ground.